Convolutional Neural Network

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Table of Contents

Convolutional Neural Network

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Convolutional Neural Network

- Convolutional neural network (CNN) is one of feed-forward neural networks.
- It usually solves a classification task.
- It can be composed of convolutional layers, pooling layers, and fully-connected layers.
- An activation function, which is a non-linear transformation can be applied in a layer.
- Usually, three-dimensional data (for multi-channel image case) is fed into an input layer, and the last layer produces an output as one-hot representation (for classification task).

Convolutional Neural Network

- A regular three-layer neural network.

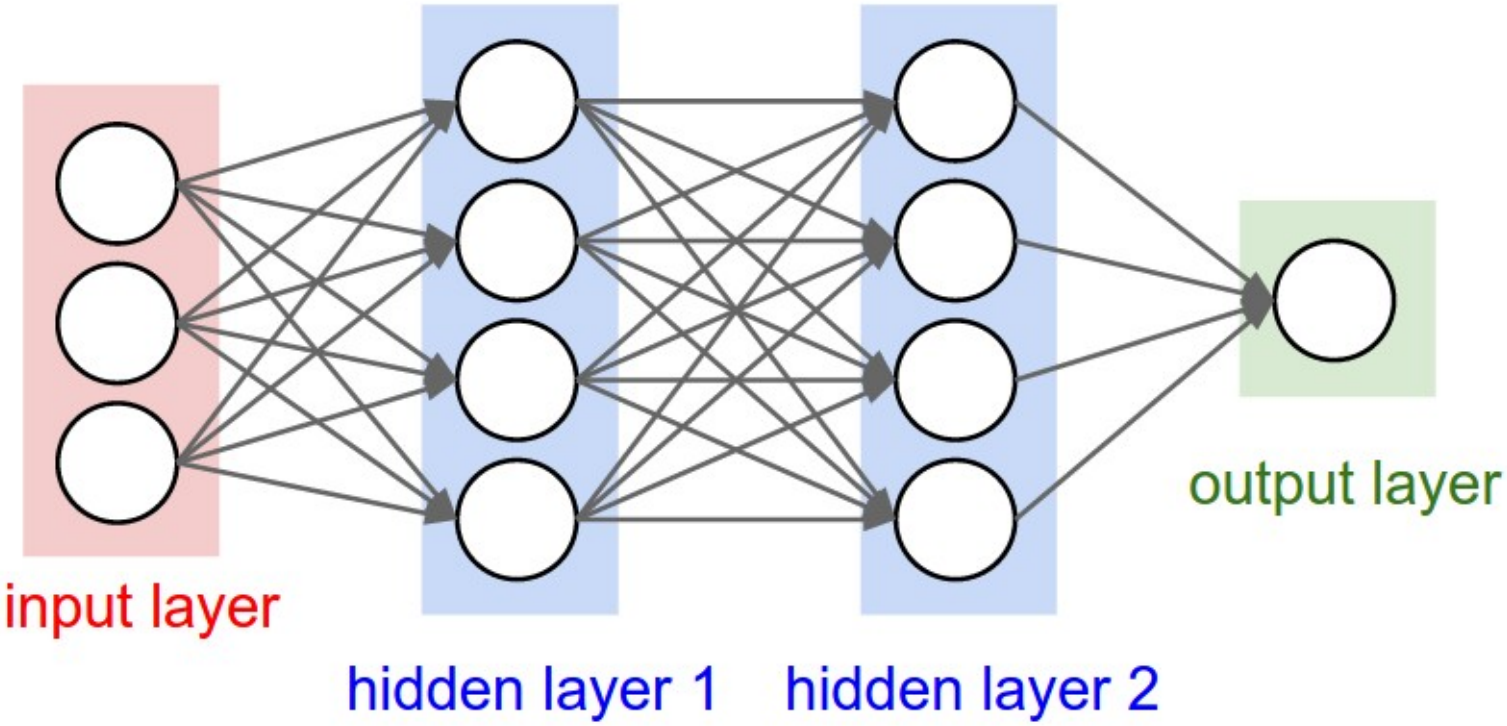

Figure 1: http://cs231n.github.io/convolutional-networks/

Convolutional Neural Network

- CNN is built with many neurons in three dimensions.

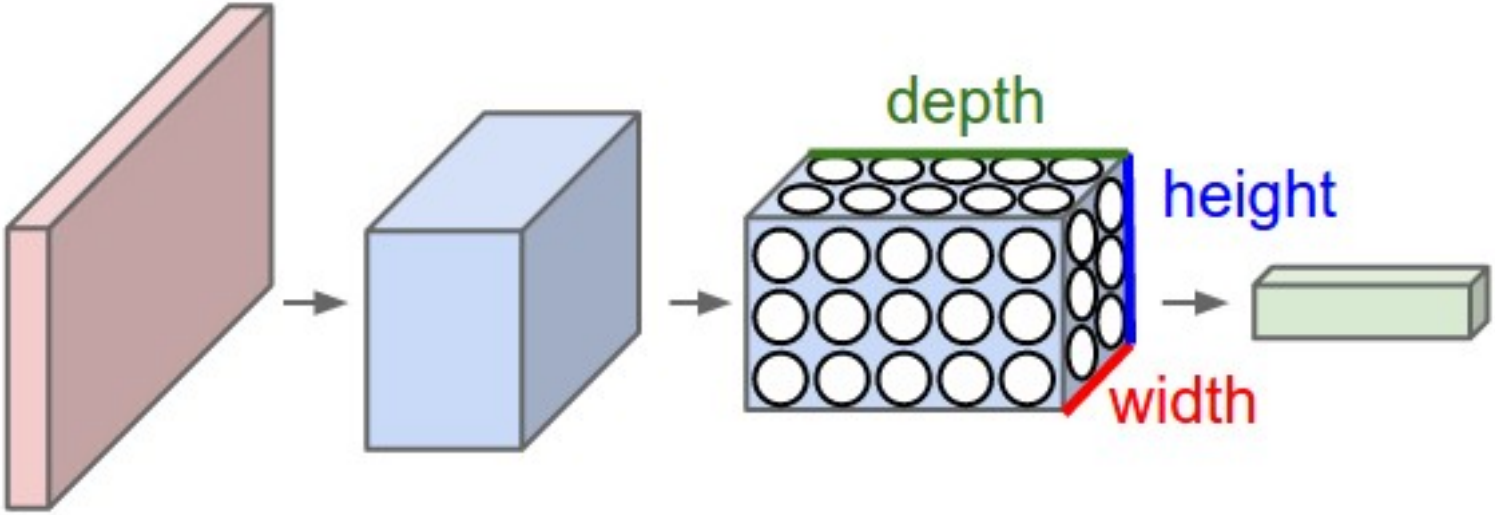

Figure 2: http://cs231n.github.io/convolutional-networks/

Architecture of Convolutional Neural Network

- From now, the components of CNN is introduced in detail.
- Components of CNN for classification task:
  - input layer
  - convolutional layer
  - pooling layer
  - fully-connected layer
  - output layer
  - activation fuction
  - cross entropy.

Input Layer

- In this slides, we use an image dataset as training and test datasets.

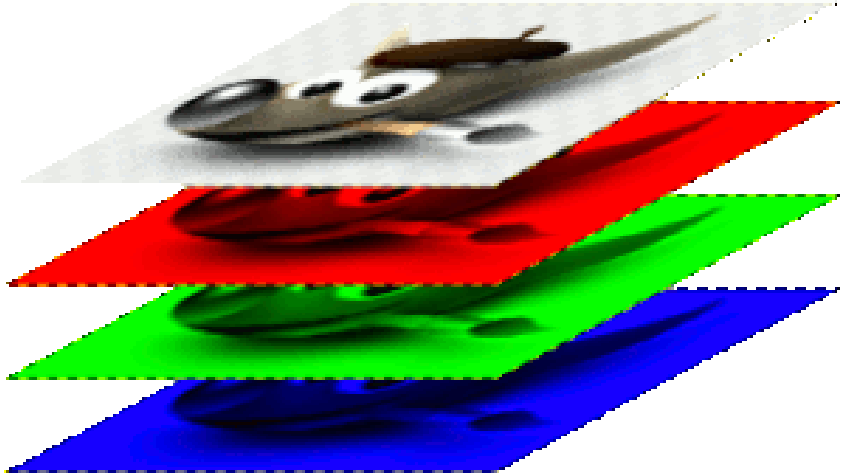

Figure 3: https://docs.gimp.org/2.4/en/gimp-images-in.html

Convolutional Layer

- Convolutional layer consists of a set of learnable filters (or kernels).
- It usually has parameters as four-dimensional tensor, `(kernel width, kernel height, input channel, output channel)`.
- It is known to capture local features in images.

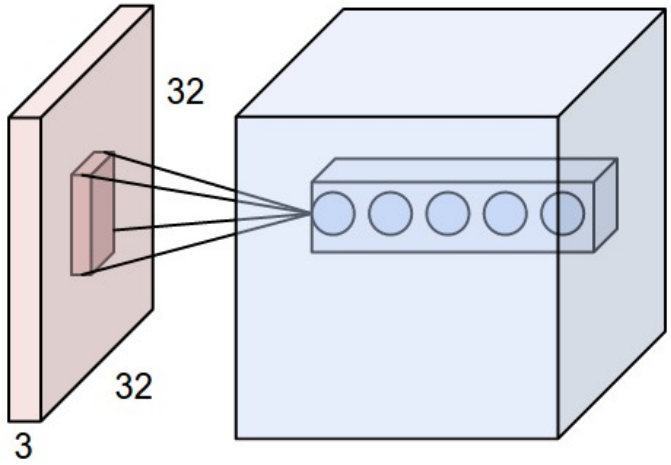

Figure 4: `http://cs231n.github.io/convolutional-networks/`

Convolutional Layer

- It reduces the number of parameters, connecting neurons locally and sharing their parameters.
- To apply the filters, `stride` and `zero-padding` should be set.

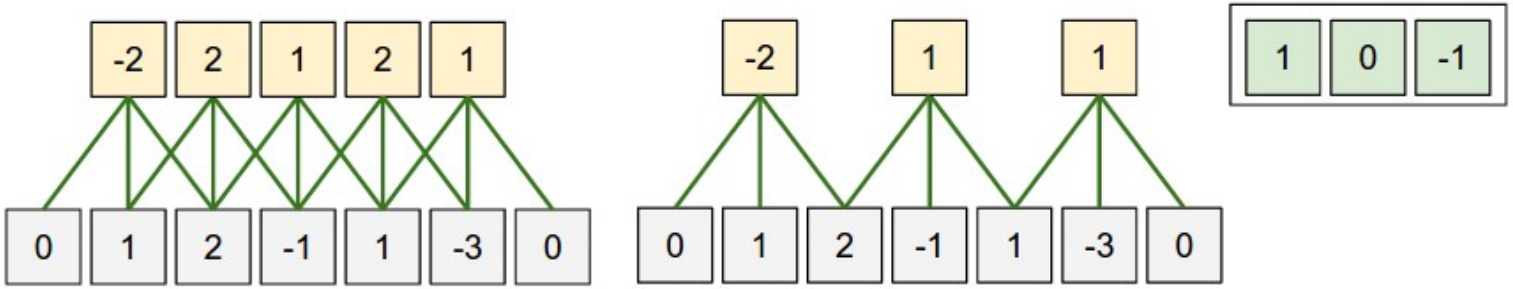

Figure 5: `http://cs231n.github.io/convolutional-networks/`

Pooling Layer

- Pooling layer downsamples an input tensor with respect to width and height.
- It can reduce the number of parameters and control overfitting.

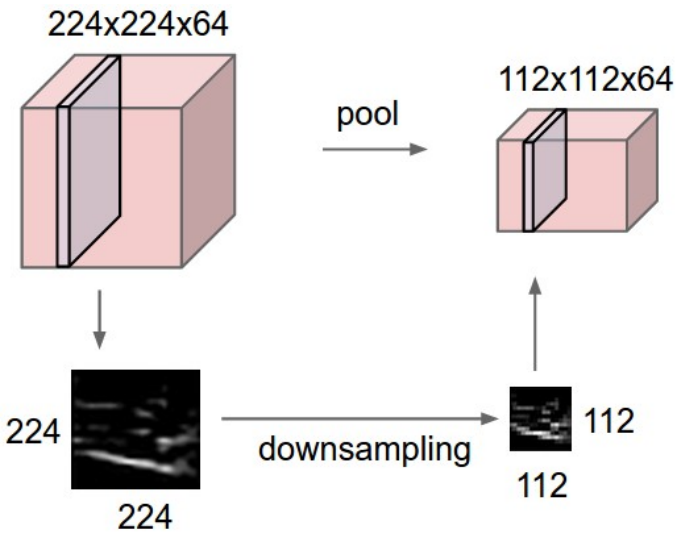

Figure 6: `http://cs231n.github.io/convolutional-networks/`

Pooling Layer

- There are several methods to downsample, such as max pooling and average pooling.
- For back-propagation, the indices of downsampled nodes are kept during forward-propagation.

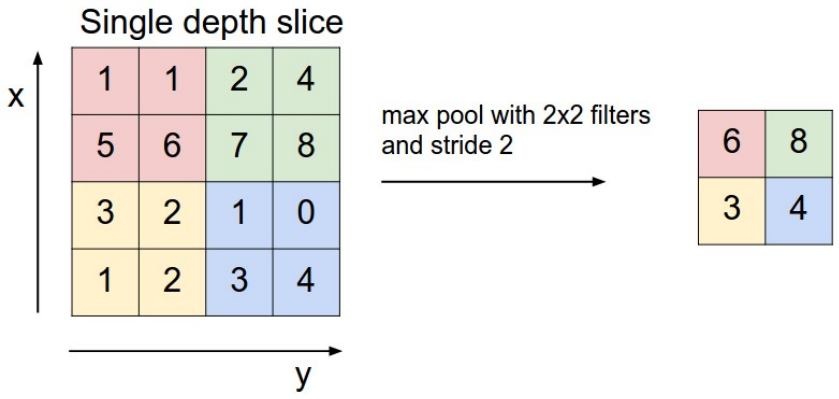

Figure 7: `http://cs231n.github.io/convolutional-networks/`

Fully-Connected Layer

- Fully-connected layer connects all nodes between two layers.
- It can be written as

$$y = \mathbf{w}^\top \mathbf{x} + b.$$

Output Layer

- For classification task, the dimension of output layer is the number of classes.
- Class probabilities of each data can be computed by softmax function:

$$p(\mathbf{z}) = [p(z_1) \cdots p(z_k)]^\top$$

where

$$p(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^{k} \exp(z_j)}$$

for $1 \leq i \leq k$.

Activation Function

- It is a function to express the switch which has two outputs, ON and OFF.
- In a neural network field, it is non-linear and its shape is usually sigmoid.
- There are several activations such as logistic function, hyperbolic tangent function, and rectified linear unit.

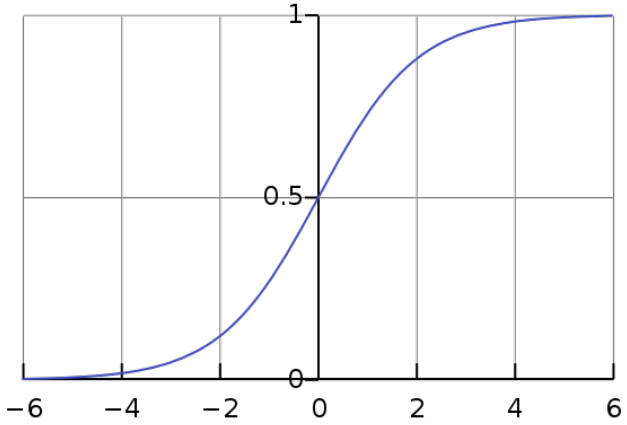

Figure 8: From Wikipedia

Cross Entropy

- Cross entropy is usually used as a loss function for classification task.
- The cross entropy between two probability distributions p and q is defined as

$$H(p, q) = -\sum_{\mathbf{x}} p(\mathbf{x}) \log q(\mathbf{x})$$

for discrete p and q.

Cross Entropy

- Minimizing the cross entropy is equivalent to maximizing log-likelihood:

$$\frac{1}{N}\log\prod_{i=1}^{k} q_i^{Np_i} = \sum_{i=1}^{k} p_i \log q_i = -H(p,q)$$

where N is the number of training data and k is the number of classes.

Famous Architecture of Convolutional Neural Network

- LeNet-5
- AlexNet
- VGGNet
- GoogLeNet
- Inception-v2, Inception-v3
- ResNet
- Inception-v4

- http://slazebni.cs.illinois.edu/spring17/lec01_cnn_architectures.pdf

References

- Professor Seungjin Choi's materials
- http://cs231n.github.io/convolutional-networks/
- Wikipedia

Thank you.